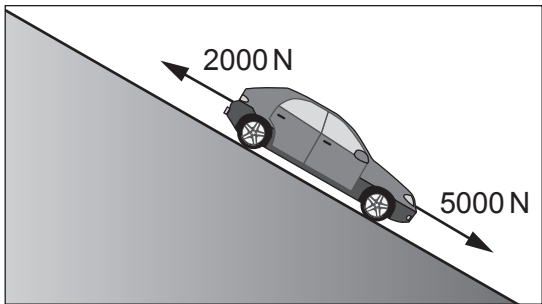


2. The diagram shows a car rolling down a slope. Two of the forces acting on the car are labelled.



(a) The car has a **weight** of 10 000 N. Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the car.
(Gravitational field strength, g , = 10 N/kg)

[2]

mass = kg

(b) Use the information in the diagram to answer the following questions.

(i) Calculate the resultant force acting down the slope.

[2]

resultant force = N

(ii) Use your answers from parts (a) and (b)(i) and the equation:

$$\text{acceleration} = \frac{\text{resultant force}}{\text{mass}}$$

to calculate the acceleration of the car at the instant shown and state the unit. [3]

acceleration =

unit =



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only

(iii) I. Explain how the resultant force on the car changes as it speeds up. [2]

.....

.....

.....

II. State how this change in resultant force affects the acceleration of the car. [1]

.....

(c) At the bottom of the slope the car continues horizontally at a constant speed of 12 m/s with a kinetic energy of 72 000 J.

(i) State **one** reason why the potential energy at the top of the hill must have been greater than 72 000 J. [1]

.....

(ii) At the bottom of the hill a braking force is applied which stops the car over a distance of 15 m.

Use the equation:

$$\text{force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the braking force. [2]

braking force = N

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